

Michael Mulugeta Demissie

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SUMMARY

Software Engineer with 4+ years building and debugging production-level backend systems, APIs, and AI-powered applications, now focused on evaluating and improving large language models on technical work. Skilled in rubric-based evaluation and ranking of AI-generated responses to engineering prompts for correctness, efficiency, and clarity, and in authoring training data and structured technical feedback. Experienced collaborating asynchronously with cross-functional teams. Strong attention to detail, guideline compliance, and written communication. Domain: Software Engineering and NLP.

EDUCATION

The George Washington University — M.S. Data Analytics

Expected May 2027

Delhi Technological University — B.Tech Software Engineering

Jul 2025

PUBLICATIONS

Cost-Effective Real-Time Targeted Advertising on Public Screens — Springer (SCI 2025 proceedings)

Jan 2026

- Co-authored Springer paper on a real-time targeted-advertising framework using CNN-based gender and age detection with live weather data to personalize public-screen ads.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, Bash, Node.js

Software Engineering: Production backend systems, REST APIs, system design, microservices, code review, debugging, full-stack development, third-party API integration

AI Training: LLM evaluation, rubric-based scoring, response ranking and comparison, data annotation, training-data authoring, structured feedback, prompt engineering, guideline compliance, bias awareness

AI and ML: OpenAI API, HuggingFace, BERT, DistilBERT, LightGBM, PyTorch, scikit-learn, RAG, pgvector, Groq, NLP

Cloud and Infra: AWS (S3, Lambda), Docker, CI/CD, Railway, Render

Databases: PostgreSQL, MySQL, Redis

Frameworks: FastAPI, Flask, Express, React, Airflow, Pandas, NumPy, Git

EXPERIENCE

Expert Software Developer | micro1 · Remote

May 2026 to Present

- Execute and test production LLMs on software-engineering tasks, recording structured observations on whether each model succeeded, met requirements, and how long it took to run.
- Run models across multiple rounds to probe edge cases, document failures, and submit evaluated outputs with detailed written observations.
- Evaluate and rank model responses for correctness, efficiency, and clarity against detailed technical guidelines.
- Identify and report issues to assigned project specialists, collaborating with the team to improve task quality.

AI Trainer – Software Engineering Expert | Handshake AI Fellowship · Remote

May 2026 to Present

- Developed and evaluated domain-specific prompts to assess the performance of large language models (LLMs) on software-engineering tasks.
- Analyzed and ranked LLM outputs for correctness, efficiency, and clarity using rubric-based scoring.
- Authored training data, including code solutions and structured instructions that guide model behavior on technical tasks.
- Contributed to agentic AI evaluation, assessing model performance on multi-step, real-world computer-use tasks.
- Delivered structured written feedback under client guidelines and strict confidentiality requirements.

Software Engineer | Hasab AI · Remote

June 2024 to June 2025

FastAPI · PostgreSQL · Redis · Docker · AWS

- Built and maintained the API layer for a voice-intelligence platform delivering speech-to-text, translation, and text-to-speech for low-resource African languages.
- Developed and debugged production backend services, designing endpoints and architecture to support transcription and language-processing features.
- Collaborated with the team to integrate AI-powered speech services into the platform.

Research Assistant | Delhi Technological University — MeitY-Sponsored AI Project

Dec 2024 to May 2025

Python · HuggingFace · PyTorch

- Trained and fine-tuned language models on a ~2M-row text dataset as part of a government-sponsored AI research project.
- Designed and ran experiments that improved model F1 score from 79% to 88% through iterative tuning and analysis.
- Built and curated the training dataset and contributed to project deliverables alongside the research team.

Software Engineer Intern | ParallelDots

June 2024 to Nov 2024

Node.js · Express · PostgreSQL · MySQL

- Built and debugged client-facing REST API endpoints for an AI computer-vision platform serving Fortune 500 clients.
- Integrated third-party services into the backend and supported image-processing API features following engineering best practices.

Freelance Software Engineer | Self-Employed · Remote

Feb 2022 to Present

Python (FastAPI/Flask) · Node.js · Express · React · PostgreSQL · MySQL

- Delivered 10+ full-stack projects for startups, small businesses, and individual clients sourced through freelance platforms, building backend APIs, web applications, and data/automation scripts.
- Completed 100+ assistive engagements, including bug fixes, feature additions, and partial builds on existing codebases.
- Designed and deployed REST APIs and database-backed services, handling backend logic and React frontends end to end while managing scope and delivery independently.

TECHNICAL PROJECTS

PromptBench: LLM Response Evaluation Framework | Software Engineer and AI Developer

Oct 2025 to Feb 2026

Python · FastAPI · HuggingFace · OpenAI API · PostgreSQL · Docker

- Built a production-level LLM evaluation framework generating 400+ domain-specific technical prompts and ranking responses across 4 models using rubric-based scoring for correctness, efficiency, and clarity.
- Designed the PostgreSQL schema to store prompts, model responses, and scores, enabling structured comparison and analysis across evaluation runs.

DocuQuery: Production RAG Pipeline and AI Document Assistant | Software Engineer and AI Developer

Apr 2026 to Jun 2026

Python · FastAPI · PostgreSQL · pgvector · HuggingFace · Groq LLM · Airflow · Docker · Railway · [Live Demo](#)

- Built and deployed a production-level RAG pipeline serving streaming LLM responses to natural-language questions, with response evaluation for correctness and clarity.
- Orchestrated document ingestion and embedding with Airflow, storing vector embeddings in pgvector for semantic retrieval.

FinResolve: AI-Powered Financial Complaint Routing | Full Stack and ML Engineer

Jan 2026 to May 2026

Python · FastAPI · DistilBERT · LightGBM · PostgreSQL · AWS S3 · Docker · Render · [Live Demo](#)

- Built and deployed a production-level system using fine-tuned DistilBERT and LightGBM, reaching 88% accuracy on 1M+ real-world NLP records.
- Fine-tuned DistilBERT to classify and route financial complaints to the correct category, with LightGBM supporting prediction and architecture following engineering best practices.

Distributed Microservices Pipeline | Backend and Data Engineer

May 2026 to Jun 2026

Node.js · PostgreSQL · Airflow · Docker · Express · Railway · WMATA API

- Designed and built a production-level real-time backend processing 580,000+ rows/day across five containerized microservices, architected for correctness, efficiency, and reliability.
- Ingested real-time transit data from the WMATA API and orchestrated the five services with Airflow to keep data flowing on schedule.